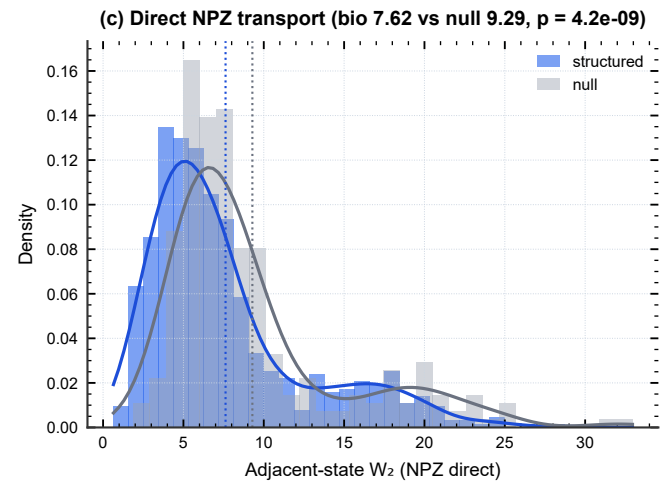
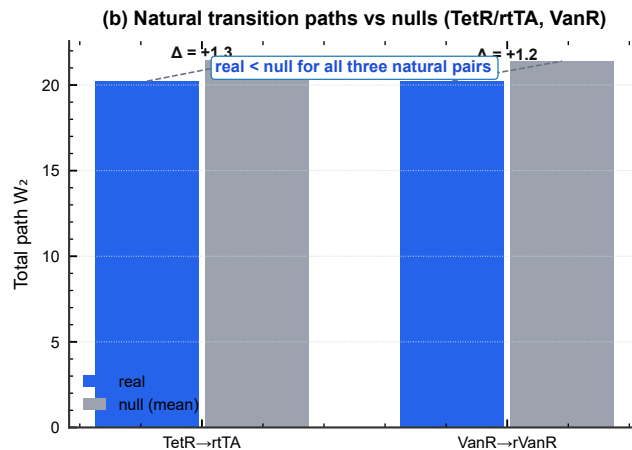
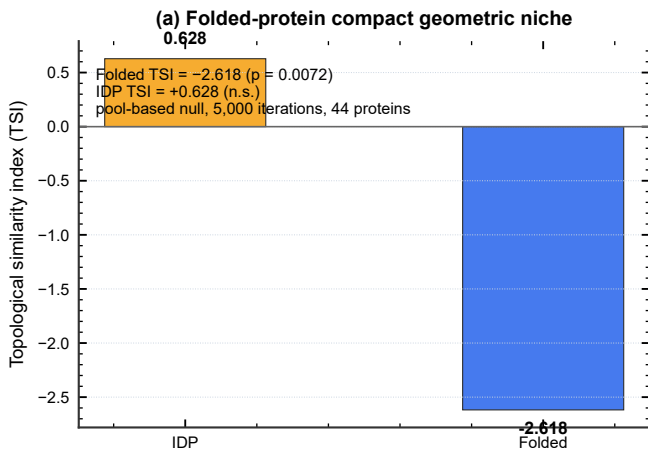




(d) Scope of the validated claim (main-text Fig. 4d)

Validated: transport component (W_2)

structured paths < matched null paths
 across joint-PCA, heteropolymer and direct-NPZ analyses

primary: 11.75 vs 22.90, $d = -1.54$, $p = 8.4 \times 10^{-68}$

hetero 3.07 vs 3.46 ($d = -0.20$) · NPZ 7.62 vs 9.29

Theoretical extensions (not validated here)

$A[y] = \bar{W}_2[y] + \text{reorg}[y] + \text{feas}[y]$

reorganization and feasibility terms are retained as theory,
 not as experimentally established physical action

transition-sharpness is not independent evidence
 for a global action minimum